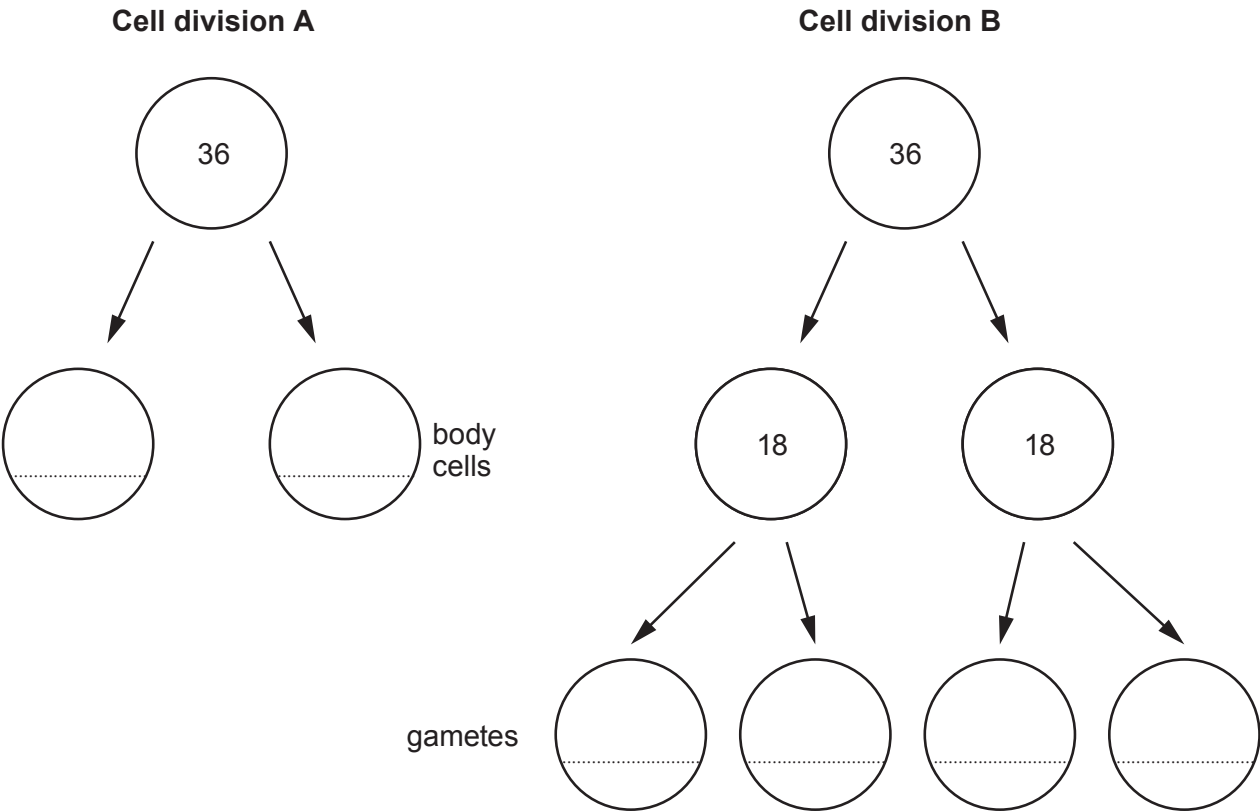


5. Chromosomes are structures made of DNA and are found inside the nucleus of cells. The meerkat (*Suricata suricatta*) has 36 chromosomes in its body cells.



(a) **Image 5.1** shows two different types of cell division in the **meerkat**.

Image 5.1



- (i) **Complete Image 5.1** by writing in the number of chromosomes in **each** body cell and gamete. [2]
- (ii) State the name of cell division **A**. [1]

.....

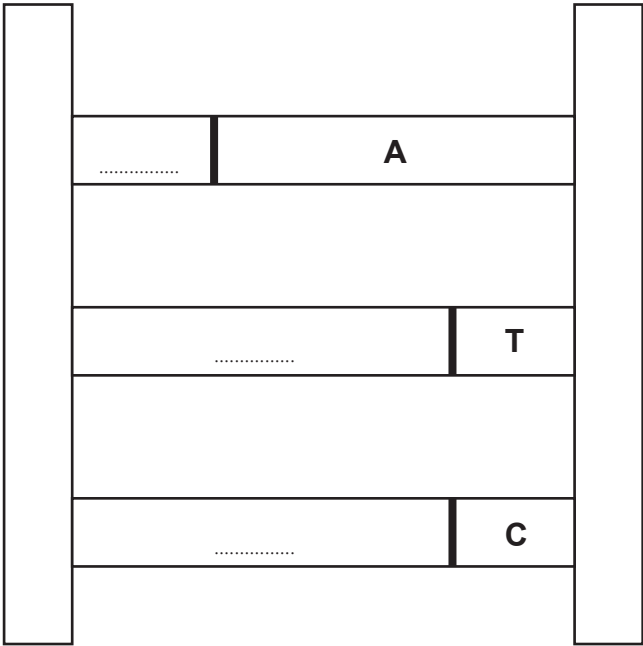


(iii) State the disease that occurs when cell division **A** is uncontrolled. [1]

.....

(b) **Image 5.2** shows part of a DNA molecule. Complete **Image 5.2** by adding the correct letters to represent the names of the missing bases. [2]

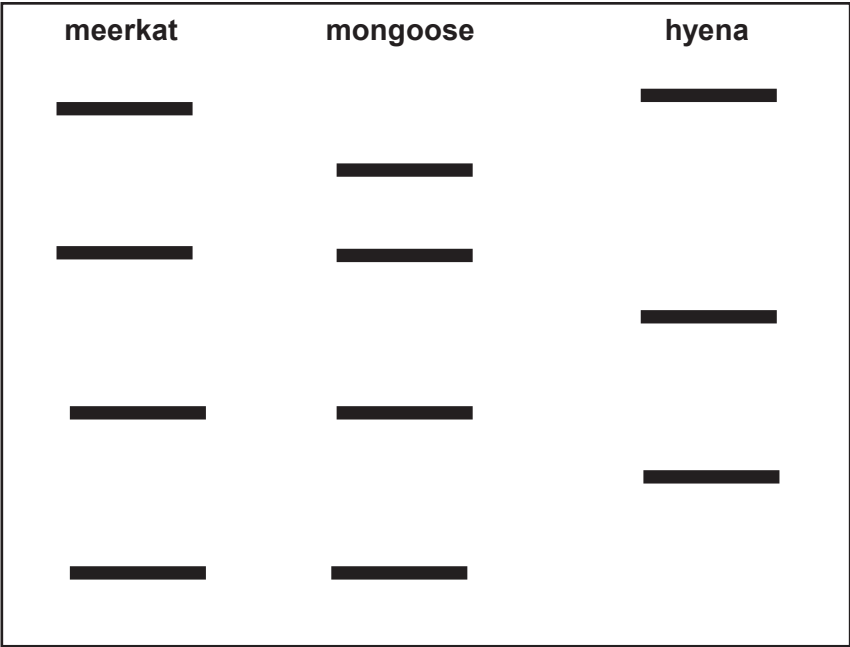
Image 5.2



Examiner
only

- (c) Scientists use DNA profiling to find out how closely related different species of animals are. The DNA profiles of three different animals are shown in **Image 5.3**.

Image 5.3



Use the DNA profiles in **Image 5.3** and your knowledge to answer the following questions:

- (i) State the **two** animals which are most closely related. Explain your answer. [2]

Animals

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- (ii) State **one** other use for DNA profiling. [1]

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